✦ CONFIDENTIAL WHITEPAPER ✦

S P A C E   A I

Zero-Knowledge Blockchain Infrastructure for Space Data & AI Automation

Version 1.0 · 2026
Built on Aleo ZK-Proof Architecture

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The global space economy is projected to surpass $1.8 trillion by 2035, yet its data infrastructure remains fragmented, opaque, and inaccessible to decentralized capital markets. Simultaneously, AI-driven automation systems demand real-time, verifiable, and privacy-preserving data pipelines — a need that legacy cloud architectures cannot satisfy.

SPAI (Space AI) is a next-generation blockchain ecosystem that unifies satellite-derived data processing, AI autonomous computation, and zero-knowledge cryptography into a single programmable settlement layer. Built upon the Aleo privacy blockchain, SPAI enables enterprises and developers to transact with verified space data — securely, privately, and at global scale.

The SPAI token serves as the native utility and deflationary reserve asset of this ecosystem. Unlike inflationary token models that have eroded investor confidence across the industry, SPAI employs a fully-diluted launch structure combined with a protocol-level buyback-and-burn mechanism, ensuring that ecosystem growth directly reduces circulating supply over time.

> **Core thesis: Space data is the most undermonetized asset class of the 21st century. SPAI transforms it into a programmable, verifiable, and tradeable digital resource.**

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2.1 The Trillion-Dollar Fragmentation Problem

The space infrastructure sector suffers from three critical structural failures that prevent it from realizing its full economic potential:

- Data Siloing: Satellite operators, government agencies, and commercial providers maintain closed data ecosystems. Cross-operator data verification is costly, slow, and often technically impossible. A logistics firm seeking multi-constellation GPS validation must contract with 4–7 separate vendors.
- Payment Infrastructure Gap: There is no universal settlement layer for space data commerce. Transactions are processed through traditional wire transfers with 30–90 day settlement windows, introducing counterparty risk and limiting market participation to well-capitalized institutions.
- AI Compute Trust Deficit: As AI systems increasingly rely on satellite data for critical decisions (autonomous vehicles, agricultural optimization, disaster response), there is no cryptographic mechanism to verify that AI computations were performed on authentic, unmanipulated data feeds.

2.2 Market Scale

Market Segment	2024 Value	2035 Projection	CAGR
Global Space Economy	$630B	$1.8T	+10.2%
Satellite Data Services	$8.4B	$36B	+14.7%
AI Automation (Industrial)	$190B	$826B	+13.9%
Global Logistics (Digital)	$22B	$98B	+14.5%

2.3 Why Blockchain is the Missing Layer

Blockchain infrastructure solves all three structural failures simultaneously: it enables permissionless data settlement, cryptographic provenance verification, and programmable payment automation — without requiring trust in any single operator.

Zero-knowledge proofs extend this capability further: computations can be verified as correct without revealing the underlying data, enabling privacy-preserving AI inference on sensitive satellite streams.

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3.1 The SPAI Protocol Stack

SPAI introduces a four-layer architecture that transforms raw satellite data into programmable economic primitives:

Layer 1 — Data Ingestion & Verification

Satellite data streams are ingested through SPAI Oracle Nodes — a distributed network of validators that cryptographically attest to data authenticity. Each data packet is assigned a zero-knowledge proof of provenance before entering the SPAI mempool.

Layer 2 — AI Compute Layer

Licensed AI inference engines operate within SPAI's zkVM (zero-knowledge virtual machine), inherited from Aleo's Leo compiler architecture. Computation results are published on-chain with a verifiable proof — enabling any third party to confirm correctness without seeing the raw inputs.

Layer 3 — Settlement & Tokenization

SPAI tokens serve as the universal settlement currency for all data transactions, AI compute fees, and infrastructure licensing within the ecosystem. Smart contracts automate revenue distribution, buyback triggers, and staking rewards without human intermediation.

Layer 4 — Application Gateway

Enterprise clients access SPAI data APIs through a permissioned gateway that converts on-chain verified data into standard industry formats (REST, gRPC, WebSocket). This layer enables integration with existing ERP, logistics, and financial infrastructure.

3.2 Primary Use Cases — Phase 1 Focus

Phase 1 of SPAI concentrates exclusively on a single high-value use case to establish product-market fit before expanding:

Phase 1 Core Use Case: Verified Satellite Data Settlement for Commercial AI Applications

- Agricultural Intelligence: AI crop monitoring systems subscribe to verified multispectral satellite data streams via SPAI, paying in SPAI tokens per data epoch.
- Maritime & Logistics Tracking: Shipping operators access real-time AIS and SAR satellite data with cryptographic proof of accuracy for insurance and compliance purposes.
- Disaster Response Networks: NGOs and government agencies access priority satellite imagery feeds through SPAI's emergency data protocol.

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4.1 Aleo ZK-Proof Foundation

SPAI is built natively on the Aleo blockchain — the world's first Layer 1 purpose-built for zero-knowledge applications. This architectural choice provides three irreplaceable capabilities:

Capability	Aleo Implementation	SPAI Benefit
Private Computation	zkSNARK via Leo Compiler	AI inference on private data feeds
Programmable Privacy	Record-based ownership model	Selective data disclosure for enterprise clients
Proof Verification	Marlin proof system	Trustless on-chain validation of satellite data provenance
Smart Contracts	Leo programming language	Automated settlement, buyback, and staking logic

4.2 SPAI-Specific Protocol Extensions

Oracle Network — SpaceNode

SpaceNodes are a new validator class within the SPAI network responsible for satellite data ingestion. SpaceNode operators must stake a minimum of 100,000 SPAI tokens and maintain verifiable uptime SLAs. Rewards are distributed from the Protocol Fee Pool (see Tokenomics Section 5.3).

zkAI Module

The zkAI Module is a SPAI-developed extension to Aleo's Leo VM that enables AI model inference within zero-knowledge circuits. Model weights are committed on-chain as Pedersen hash commitments; inference results are accompanied by SNARK proofs that verify execution against the committed model.

Phase 2 Architecture — SPAI L2 Rollup (Roadmap Item)

To support high-throughput data commerce at enterprise scale, Phase 2 will introduce a SPAI-native ZK rollup layer. Transaction batching will reduce on-chain footprint by an estimated 95%, lowering per-transaction costs while maintaining Aleo's security guarantees as the settlement layer.

4.3 Security Model

- All smart contract code is open-source and subject to mandatory third-party audit prior to mainnet launch.

- Oracle data feeds require a supermajority (2/3+) of SpaceNode validators to sign each data epoch before on-chain submission.
- Emergency pause functionality is controlled by a 5-of-9 multi-signature council with geographically distributed keyholders.

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5.1 Token Overview

Parameter	Value
Token Name	SPAI (Space AI Token)
Token Standard	Aleo Native Asset (Leo Record Type)
Maximum Supply	1,000,000,000 SPAI (Hard Cap — No Additional Minting)
Launch Circulating Supply	~720,000,000 SPAI (72% — Fully Diluted at TGE)
Remaining Supply	280,000,000 SPAI (Linear Vesting — Team, Advisors, Reserve)
Inflation Rate	0% — Fixed Supply Forever
Deflationary Mechanism	Buyback & Burn from Protocol Revenue

5.2 Distribution Architecture

SPAI's token distribution is designed around a core principle: the market receives the vast majority of tokens at launch, eliminating the overhang risk that has destroyed investor confidence in previous projects.

Allocation	Amount	% of Supply	Vesting Schedule
Public Sale & DEX Liquidity	350,000,000	35%	100% unlocked at TGE
Ecosystem & Developer Grants	200,000,000	20%	100% unlocked at TGE (locked in protocol DAO)
Community Rewards Pool	170,000,000	17%	100% unlocked at TGE (distributed via staking)
Team & Core Contributors	120,000,000	12%	12-month cliff → 36-month linear vesting
Strategic Investors (Seed/A)	80,000,000	8%	6-month cliff → 24-month linear vesting
Foundation Reserve	60,000,000	6%	24-month cliff → 24-month linear vesting
Advisors	20,000,000	2%	12-month cliff → 24-month linear vesting

KEY DESIGN PRINCIPLE: 72% of supply is market-accessible at TGE. The remaining 28% is subject to hard-coded vesting contracts on Aleo — early unlocks are cryptographically impossible, not merely contractual promises.

5.3 Revenue Model & Buyback-Burn Engine

The protocol generates revenue through four streams. All protocol revenue flows into the SPAI Treasury smart contract, which automatically executes buyback-and-burn operations on a monthly epoch basis:

Revenue Stream	Mechanism	Buyback Allocation
Data API Subscriptions	Enterprise clients pay monthly USDC/USDT fees for data access	40% of revenue → buyback
AI Compute Fees	zkAI module charges per-inference fees in SPAI tokens	30% burned directly (no buyback needed)
Network Gas Fees	All SPAI network transactions pay gas in SPAI	50% burned, 50% to SpaceNode rewards
Data Licensing	One-time licensing fees for proprietary satellite datasets	60% of revenue → buyback

Buyback Mechanics

On the first block of each monthly epoch, the Treasury contract reads the accumulated USDC/USDT balance, executes a market buy of SPAI through the protocol's designated DEX liquidity pools, and immediately sends purchased tokens to the zero address (burn). All transactions are publicly verifiable on-chain.

5.4 Staking & Node Rewards — Zero Inflation Model

Staking rewards are funded entirely from protocol revenue — not from new token minting. This distinction is critical: SPAI stakers are receiving a share of real economic activity, not diluted inflation subsidies.

Reward Source	Distribution Target	Notes
50% of Gas Fee Burns	SpaceNode Operators	Proportional to staked SPAI + uptime score
10% of Buyback Pool	SPAI Stakers (non-node)	Distributed before burn execution
Ecosystem Grants Pool	Developer Incentives	DAO-governed quarterly grants

Target APY for SpaceNode operators: 12–18% (variable, tied to network activity). Target APY for general stakers: 4–8%. These figures are projections based on modeled network usage and will vary with actual protocol revenue.

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Phase 1 — Foundation (Q3 2026 – Q1 2027)

Q3 2026 — Protocol Bootstrap

- Leo smart contract development: Token issuance, vesting, buyback-burn contracts
- SpaceNode alpha network launch (20 permissioned validator nodes)
- First enterprise partnership signed: maritime logistics data pilot
- Security audit engagement (Halborn / Trail of Bits)

Q4 2026 — TGE & Market Launch

- Token Generation Event (TGE) — 72% circulating supply at launch
- DEX liquidity deployment on Aleo-native and cross-chain bridges
- SPAI Data API v1.0 public beta (3 satellite data providers integrated)
- Community staking program launch

Q1 2027 — Product-Market Fit

- 10 enterprise clients onboarded (target: $500K ARR)
- First monthly buyback-burn executed and publicly verified
- zkAI Module testnet launch — initial AI inference use cases
- SpaceNode network expanded to 100 permissionless validators

Phase 2 — Expansion (Q2 2027 – Q4 2027)

- SPAI L2 ZK Rollup mainnet launch — 95% transaction cost reduction
- Agricultural and disaster response vertical expansion
- Cross-chain bridge deployment (Ethereum, Solana, Cosmos IBC)
- zkAI Module mainnet: First production AI inference proofs
- $5M ARR milestone target — protocol buyback pressure meaningful
- DAO governance activation — community control of treasury allocation

Phase 3 — Global Infrastructure (2028+)

- Physical SpaceNode hardware program — purpose-built validator nodes for satellite ground stations
- SPAI Logistics Tokenization Module: Real-world asset tokenization of global logistics routes
- Government and institutional API tier (SOC 2 Type II compliance)
- $50M ARR target — sustained deflationary pressure via buyback engine
- Exploration of sovereign space data partnerships (national space agencies)

Milestone	Target Date	KPI
TGE & DEX Launch	Q4 2026	Circulating supply 72% at day 1
First Buyback-Burn	Q1 2027	> $100K SPAI burned in month 1
Enterprise ARR $500K	Q1 2027	10+ paying enterprise clients
L2 Rollup Mainnet	Q3 2027	< $0.001 per transaction
Protocol ARR $5M	Q4 2027	Sustained monthly buyback > $150K
Protocol ARR $50M	2028+	Top-50 DeFi protocol by fee revenue

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Prospective participants should carefully consider the following risk factors before engaging with the SPAI ecosystem:

Regulatory Risk

The regulatory environment for digital assets and tokenized infrastructure is rapidly evolving. SPAI tokens may be subject to securities classification, AML/KYC requirements, or outright prohibition in certain jurisdictions. The Foundation will pursue legal opinions in all target markets but cannot guarantee regulatory stability.

Technology Risk

Aleo's Leo VM and ZK proof systems are emerging technologies. Cryptographic vulnerabilities, protocol upgrades, or performance limitations may require significant architectural changes to the SPAI protocol.

Market Risk

Token prices are subject to extreme volatility. Buyback-burn mechanisms reduce supply over time but do not guarantee price appreciation. Enterprise revenue projections are forward-looking estimates and may not be achieved.

Execution Risk

Satellite data partnerships require complex commercial negotiations with established aerospace and data vendors. Delays in enterprise onboarding could reduce protocol revenue and buyback pressure below projected levels.

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SPAI represents a fundamentally new approach to space data monetization — one that aligns the interests of data providers, AI developers, enterprise consumers, and token holders through cryptographic primitives rather than legal contracts and trust assumptions.

By launching with 72% of supply in market circulation, SPAI eliminates the overhang dynamics that have eroded trust in token projects. By funding staking rewards from real protocol revenue rather than inflation, SPAI creates a sustainable economic loop that strengthens with adoption. By building on Aleo's zero-knowledge foundation, SPAI ensures that privacy and verifiability are architectural guarantees — not optional features.

> **The space economy is entering its commercial era. SPAI is the settlement layer that makes it programmable.**

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